Advantages	Disadvantages
The USB interface is a universal standard for connecting devices, therefore a USB port is included in many different devices.	
When a USB device to cable is inserted into a USB port, it is automatically detected. The first connection will normally involve the download of drivers to operate the hardware that has been connected. Each connection after this the driver should not need to be downloaded again.	
A USB connection can also be used to power a device, so it does not need another power source. It can also use this to charge a device, such as a mobile phone.	

Table 2.3: Advantages/disadvantages of the USB interface

Questions

12 What does USB stand for?

13 What type of data transmission method does a USB interface use?

14 What are **two** advantages of using the USB interface?

15 Can you name **five** devices that are connected using a USB interface?

16 What is the length a USB cable limited to? Can you think why it is limited to this length?

2.5 Detecting errors in data transmission

The transmission of data from one device to another is often not a perfect process. In the process of transmitting the data, interference can occur. This can cause data to be lost, data to be gained and data to change. This would not be helpful and could also lead to many issues. If a password wasn't transmitted correctly, then that person may not be able to log into their account. If a person's address wasn't transmitted correctly, they may not get email that needs to be sent to them. If a person's details are not transmitted correctly to a company, they may not get a product or service that they order. The accuracy of data is often vital, therefore there needs to be a procedure in place to detect any errors in data so that actions can be taken to correct this.

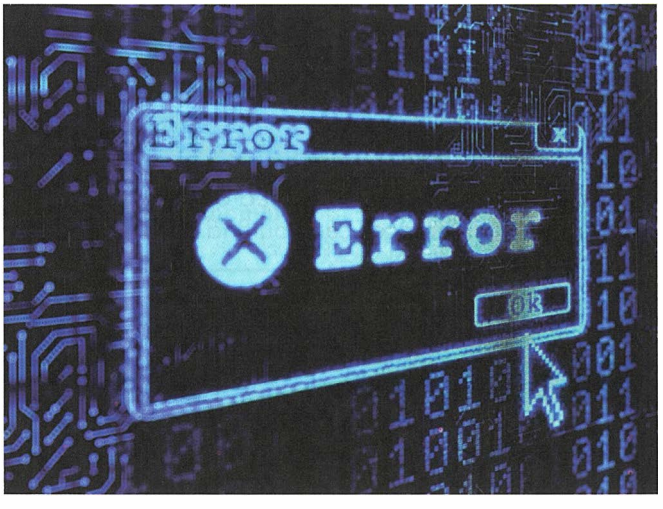


Figure 2.10: Procedures need to be in place to handle errors in data transmission

There are several error detection methods that can be used, these include:

- parity check
- checksum
- echo check.

Each of these methods is designed to check for errors after the data has been transmitted from one device to another.

Parity check

A **parity check** can use an odd or even check method. Each byte of data has 7 bits and 1 extra bit that is called a parity bit (see Figure 2.11). The parity bit is normally the first or last bit of data in the byte.

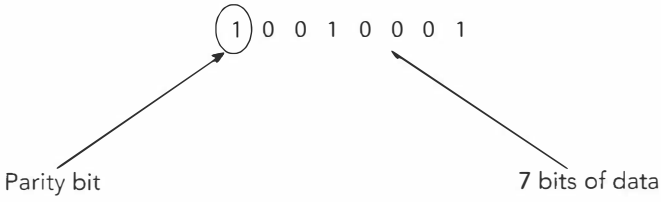


Figure 2.11: The parity bit is the first bit of data in this byte

KEY WORD

parity check: a type of error detection method that adds an additional bit to each byte to create an odd or even sum.

Before transmission begins, the parity check is set to be either odd or even parity. The number of 1s in the 7 bits of data is totalled. In the example in Figure 2.11, the result of that would be 2. If an odd parity check is used, then a 1 is added as a parity bit. This is because all the 1s in the byte now add up to 3, which is an odd number. If an even parity check is used a parity bit of 0 would have been added instead. This is because all the 1s in the byte would then add up to 2, which is an even number. When the parity bit has been added to each byte, the data can be transmitted. After transmission, the receiving device will check each byte of data for errors. If an odd parity check has been used and the device finds a byte that has an even number of 1s, then it knows that an error has occurred with this byte of data. The error is detected!

Questions

17 Which of these bytes would have been transmitted incorrectly if an even parity check has been used?

- **a** 10111011
- **b** 01110111
- **c** 10101000

18 Which of these bytes would have been transmitted incorrectly if an odd parity check has been used?

- **a** 00110001
- **b** 10110101
- **c** 10001000

19 Can you think of an instance in which an error could occur in the data, but this would not be detected by a parity check?

Checksum

A **checksum** uses a calculated value to check for errors. A value is calculated from the data that will be transmitted, before transmission takes place. A method such as modulus 11 could be used to calculate the value.

KEY WORD

checksum: a type of error detection method that performs a calculation on the data to create a checksum value. Checksum values are compared after transmission to see if they match.

ACTIVITY 2.7

Use the internet to find out how the modulus 11 method calculates a value from the data. You do not need to be able to remember how to do this method of calculation for the exam, but it will help you to know how a checksum is calculated.

Once the checksum value has been calculated it is added to the data to be transmitted with it. After transmission, the receiving device uses the same method to calculate a value from the received data. If the values match, then the device knows that no error has occurred during transmission. If the values do not match, the device knows that an error has occurred during transmission. The error is detected!

Questions

20 What is the checksum value calculated from?

21 What happens to the checksum value when it has been calculated before transmission?

22 Why would checksum values that do not match show that an error has occurred?

Echo check

An **echo check** involves a simple comparison of the data sent to the data received. The sending device transmits the data to the receiving device. The receiving device then transmits the data it receives back to the sending device. The sending device compares the data it sent to the data it has received back from the receiving device to see if they match. If they do, then no error has occurred. If they don't match, then the sending device knows the data was received with error. The error is detected!

KEY WORD

echo check: a type of error detection method that sends a copy of the transmitted data back to the sender to be compared with the original data sent.

ACTIVITY 2.8

Draw a diagram to represent how an echo check detects errors in data.

Question

23 If the data does not match that which is sent back by the receiving device, will this always mean that the receiving device did not receive the data correctly? How do you know that?

Automatic repeat request (ARQ)

When an error has been detected after the data is transmitted, it is likely that the data will need to be retransmitted. Either the sending device will need to ask the receiving device if it received the data correctly, or the receiving device will need to tell the sending device it did or did not receive the data correctly. This will allow the data to be retransmitted if necessary.

A method called **automatic repeat request (ARQ)** can be used to do this. There are two main ways that an ARQ can operate and each method uses either a positive or negative **acknowledgement** and a **timeout**.

In a positive acknowledgement method:

- The sending device transmits the first data packet.
- The receiving device receives the data and checks it for errors.
- Once the receiving device knows it has received the data error free, it sends a positive acknowledgement back to the sending device.
- When the sending device receives this positive acknowledgement, it knows the receiving device has received the data packet error free and it sends the next data packet.
- If the sending device does not receive a positive acknowledgement within a set timeframe, a timeout occurs.
- When a timeout occurs, the sending device will resend the data packet. It will keep doing this when a timeout occurs, until it receives a positive acknowledgement, or sometimes a limit (such as 20 times) is set and when this limit is reached it will stop resending the data.

ACTIVITY 2.9

Draw a diagram to represent how an ARQ system operates using positive acknowledgement.

In a negative acknowledgement method:

- The sending device transmits the first data packet.
- The receiving device receives the data packet and checks it for errors.
- If the receiving device detects no errors, no further action is taken.
- If the receiving device does detect errors, it will send a negative acknowledgement back to the sender.
- If the sender receives a negative acknowledgement, it knows this means the data was received incorrectly, so it can resend the data packet.
- A timeout is set by the sending device when it sends the data. This is just so that the sending device knows that if it doesn't receive a negative acknowledgement back within that set time period, it doesn't need to be still be waiting for it and can send the next data packet.

KEY WORDS

automatic repeat request (ARQ): a type of error detection method that uses acknowledgement and timeout to see if data has arrived correctly after transmission.

acknowledgement: a message that is sent from one device to another to indicate whether data is received correctly.

timeout: a period of time that is set and used to wait for an acknowledgement to be received.

ACTIVITY 2.10

Draw a diagram to represent how an ARQ system operates using negative acknowledgement.

REFLECTION

After completing activities 2.8, 2.9 and 2.10:
How have you made sure that your diagrams for an echo check and both ARQ methods are correct?

Check digit

You have learnt that errors, such as data loss, data gain and data being changed can occur when data is transmitted. You now know that these errors normally occur due to interference. It is also possible for errors to occur when data entry is performed. This could be manual data entry, for example, a human typing in a value, or it could be automatic data entry, for example, a barcode scanner scanning a barcode to obtain the data stored in the barcode.

A method is necessary to check for errors with data entry and this method is called a **check digit**. This is how a check digit operates:

- A check digit value is previously calculated from the data that will be entered at some point, for example, a barcode number or an ISBN number. This number is stored with the data.
- When the data is entered, for example, the barcode is scanned, the check digit is recalculated from the data entered.
- If the previously calculated check digit and the stored check digit match, the data entered is correct.
- If the previously calculated check digit and the stored check digit do not match, the data entered is incorrect.

Although the process is similar, you should not confuse the operation of a check digit and a checksum. Make sure that you take the time to understand the difference between the two.

KEY WORD

check digit: a type of error detection method that is used for data entry. A calculation is performed on the data entered to create a value. Check digit values are compared to see if the data entered is correct.

ACTIVITY 2.11

With a partner, discuss and write down the differences between a check digit and a checksum.

COMPUTER SCIENCE IN CONTEXT

Companies will normally combine several error checking methods to make sure that the data they transmit and receive is error free. This means that the data can be checked in several ways to make sure that any error that could occur is being checked. This is because the data that many companies use is vital to their business, they want to do everything they can to make sure that it does not contain any errors caused by transmitting the data.